Game Design Document

for

ArenaCraft

Version 2.0

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Revision History

Name	Date	Reason For Changes	Version
Annas, Ihsan, Roman	28.02.2026	Kickoff	1.0
Annas, Ihsan, Roman	10.04.2026	Revision	2.0

1. Game Overview

1.1 Game Concept

ArenaCraft is a 2.5D local multiplayer Battle Royale game built in Unity. Two players share a single device and compete across two distinct phases: a timed Resource & Shopping Phase, where players harvest raw materials from the arena and spend their earnings on weapons and armor upgrades, and a final Battle Royale Phase, where everyone fights with the gear they bought until only one survivor remains. The game's core tension comes from the trade-off between efficient farming and aggressive play.

1.2 Genre

ArenaCraft blends elements of the Action, Resource-Management, and Battle Royale genres. The resource-collection loop borrows from survival-crafting games while the melee combat and one-life elimination format follow the Battle Royale genre.

1.3 Target Audience

Primary audience: casual to mid-core gamers who enjoy local multiplayer with others. Players do not need prior gaming experience – the controls are simple (WASD or arrow keys) and rounds are short (4-6 min).

1.4 Game Flow Summary

All players spawn simultaneously in a shared arena. The Resource Phase begins: resource nodes (trees, stone blocks, metal deposits) are scattered around the map. Players collect resources by attacking nodes. When a player's resource bar is full or the phase timer expires, the game transitions to the Shopping Phase (1 minute) where players spend accumulated gold on weapon and armor upgrades. After shopping, the Battle Royale Phase starts. The arena is a small colosseum.

1.5 Look and Feel

The game uses a 2.5D visual style (inspired by League of Legends and Hades): 3D models and assets rendered with an isometric or orthographic camera to create depth while keeping gameplay strictly on a 2D plane (X/Z movement). The aesthetic draws inspiration from gladiatorial arenas – stone walls, torchlight, sandy floors – combined with stylized low-poly art to keep asset creation manageable for a small team. The color palette is warm earth tones (sand, terracotta, dark stone) with bright accent colors on UI elements and character highlights to ensure readability at a glance. The overall mood is competitive and energetic, with punchy sound effects and a driving percussion-based soundtrack.

2. Gameplay and Mechanics

2.1 Gameplay

2.1.1 Game Progression

Each match is self-contained and follows a fixed three-phase structure:

- *Phase 1 – Resource Phase (default: 3 minutes): Players collect resources which are converted to gold.*
- *Phase 2 – Shopping Phase (1 minute): Players spend gold on upgrades from an item shop.*
- *Phase 3 – Battle Royale Phase: Combat until one player remains.*

Progress between matches is not persistent – each round starts fresh, keeping the game balanced and replayable.

2.1.2 Mission / challenge Structure

The game is structured as individual rounds rather than a campaign. Each round is a self-contained challenge. The primary objective is to survive the Battle Royale Phase. Secondary objectives include: collecting the maximum gold before the phase ends, purchasing optimal upgrades for your playstyle, and eliminating opponents in the Battle Royale Phase to claim the win.

2.1.3 Play Flow

The rhythm of a typical round: Frantic early-game resource harvesting, with mild player interaction (players may bump into each other but combat is technically possible in Phase 1). High-intensity late-game combat in the Battle Royale Phase. Short rounds (4-6 minutes total) encourage replaying immediately.

2.2 Mechanics

2.2.1 Physics

The game uses Unity's built-in 3D physics engine. Players and objects have Rigidbody components restricted to the X/Z plane (no Y-axis movement except for cosmetic jump animations if implemented). Resource nodes are static objects with Collider components. Attack hitboxes are Trigger Colliders that detect overlapping player or node Colliders. Gravity is active but players move on a flat arena surface, so gravity is effectively invisible to the player.

2.2.2 Movement in the game

Players move on the X/Z plane using direct velocity or character controller movement. Speed is constant (no sprinting in the MVP). Controls are assigned per-player:

- *Player 1: W/A/S/D to move, Space to attack, F to interact.*
- *Player 2: Arrow Keys to move, Enter to attack, Right Shift to interact.*

2.2.3 Objects

Resource Nodes: Static objects with HP values. Players approach and press the interact/attack key to deal damage. When HP reaches 0, the node disappears and drops resource units (or auto-awards gold). Nodes respawn after a set timer or do not respawn. Item Shop Stall: A static zone activated during the Shopping Phase. Players get spawned into the zone and a UI menu appears showing available upgrades.

2.2.4 Actions

Collect: Walk up to a resource node to automatically collect it. Each press reduces node HP and awards resources. Attack: Press attack key to perform a melee swing. A short Trigger Collider is activated in front of the player for a fraction of a second. If another player is in the collider, they take damage. Shop: Spawn into the Zone and select your items from the menu. Arena: After the shopping phase spawn in the colosseum.

2.2.5 Combat

Combat uses a simple melee hit-box system. When a player presses the attack key, a Trigger Collider appears briefly in front of their character. Any opponent within that collider takes damage equal to the attacker's weapon's Damage stat. Players have a base HP of 100. Armor upgrades increase maximum HP (e.g., Light Armor = +25 HP, Heavy Armor = +50 HP, max with full armor = 150 HP). Weapon upgrades increase per-hit damage.

2.2.6 Economy

Resources are the primary collectible during Phase 1. There are diffrent resource types (e.g., Wood, Stone, Metal) with different rarities and gold values. Each resource hit on a node yields a set amount of that resource. Resources are automatically converted to gold (1 Wood = 1g, 1 Stone = 2g, 1 Metal = 5g – exact values TBD). Players cannot hold more resources than their Resource Bar capacity (default: 100 units). Gold is used in the Shop Phase to purchase upgrades:

- *Light Armor: 50 gold (+25 HP)*
- *Heavy Armor: 100 gold (+50 HP)*
- *Basic Sword: 50 gold (+moderate damage)*
- *Advanced Sword: 100 gold (+high damage)*

If the Shopping Phase timer is about to expire and the player didn't make a purchase, a warning message is displayed ('You didn't buy anything!').

2.2.7 Screen Flow

Main Menu → Player Setup Screen → Match Start (Phase 1 HUD) → Shopping Phase UI overlay → Battle Royale Phase HUD → Victory / Defeat Screen → Return to Main Menu or Rematch.

2.3 Game Options

Round Timer: Adjustable Resource Phase duration (default 3 min). Number of Players: 2. Control Scheme: WASD/Arrows with custom keybindings.

2.4 Replaying and Saving

There are no save files. Each round is a standalone session. After the Victory Screen, players can instantly rematch (same settings) or return to the Main Menu. A simple high-score or win-count tracker per session can be displayed on the Victory Screen ('P1 Wins: 3 | P2 Wins: 2').

3. Story, Setting and Character

3.1 Story and Narrative

ArenaCraft does not have a deep narrative – the game is designed for quick pick-up-and-play sessions. The implied lore is that players are gladiators in an ancient arena competing for glory and riches. No cutscenes or dialogue are planned for the MVP

3.2 Game World

3.2.1 General look and feel of world

The world is a single enclosed gladiatorial arena (Colosseum-style) rendered in a low-poly 3D art style with an isometric camera. The environment uses warm earth tones – sand, stone, terracotta – with torchlit colosseum walls and a sandy arena floor. Each player character is a low-poly humanoid gladiator distinguished by a vibrant accent color (Player 1: Red, Player 2: Blue) and starts with bare fists as their default weapon.

3.2.2 Areas

The arena is divided into informal zones: a central open area (highest resource density, highest risk of conflict), corner areas (fewer resources, safer early game). Colosseum: The final stage where the battle happens.

3.3 Characters

Players 1–2 are generic gladiator characters that are slightly different. There are low-visual background characters that synthesize a crowd.

4. Levels

4.1 Levels

There is a single level in the MVP: the Gladiator Arena. The level is a bounded rectangular or circular arena with colosseum walls preventing players from leaving. Resource nodes are distributed across the map with a mix of common (Wood), uncommon (Stone), and rare (Metal) nodes. Node placement is randomized in the MVP.

4.2 Training Level / Tutorial

The MVP does not include a dedicated tutorial level. Instead, a brief instructional overlay is shown at the start of the first match, highlighting the controls for each player, the purpose of resource nodes, and the phase structure.

5. Interface

5.1 Visual System

Each player has a dedicated UI panel positioned on the left and right side of the screen.

- *Player label (e.g., 'P1 – Red').*
- *HP Bar: Horizontal bar showing current / max HP. Turns red below 30%.*
- *Resource Bar: Vertical fill bar (fills bottom to top) showing resource capacity used.*
- *Gold Counter: Text display 'Gold: 120'.*
- *Equipped Items: Small icons for current weapon and armor.*

A shared Phase Timer is displayed at the top-center of the screen.

5.2 Control System

All controls are keyboard-based. Custom Keybindings are also available.

- *Player 1: Move – W/A/S/D | Attack – Space | Interact/Shop – F*
- *Player 2: Move – Arrow Keys | Attack – Enter | Interact/Shop – Right Shift*

5.3 Audio, music, sound effects

Background music for resourcefarming (calm nature) and battle phase (screaming, swords).

6. Technical

6.1 Target Hardware

Primary platform: PC (Windows 10/11). The game is designed to run on low-to-mid-range hardware (Intel Core i5, 8 GB RAM, integrated or entry-level GPU) to maximize accessibility.

6.2 Development hardware and software, including Game Engine

Game Engine: Unity. Programming Language: C#. Version Control: Git (GitHub). Asset Store for Assets.

6.3 Network requirements

Local play only.

7. Game Art

Simplified Art Direction
The original GDD specified a mixed art approach combining pixel art characters with low-poly 3D environments, each requiring a different toolchain. For a 3-person team this creates unnecessary overhead. The art direction has been simplified as follows:

7.1 Unified Art Style: Low-Poly 3D Only

The game uses a single consistent low-poly 3D art style for both characters and environments. This eliminates the need to maintain two separate pipelines and makes the overall visual language more coherent.

Rationale for this choice:

- Free and low-cost low-poly gladiator, character, and arena asset packs are widely available on the Unity Asset Store, dramatically reducing the art workload.
- A unified 3D pipeline (Blender → FBX → Unity) is easier to manage than juggling pixel art sprites alongside 3D environments.
- Low-poly 3D integrates naturally with Unity's lighting, camera, and physics systems, reducing setup friction.
- The isometric/orthographic camera style described in section 1.5 is a natural fit for low-poly 3D and will still achieve the intended visual depth.

7.2 Color Palette

Warm earth tones (sand, stone, terracotta) with vibrant player accent colors (Red for P1, Blue for P2) to ensure immediate readability. UI elements use high-contrast colors against the earthy background.

7.3 Asset List (Simplified)

The following assets are required for the MVP:

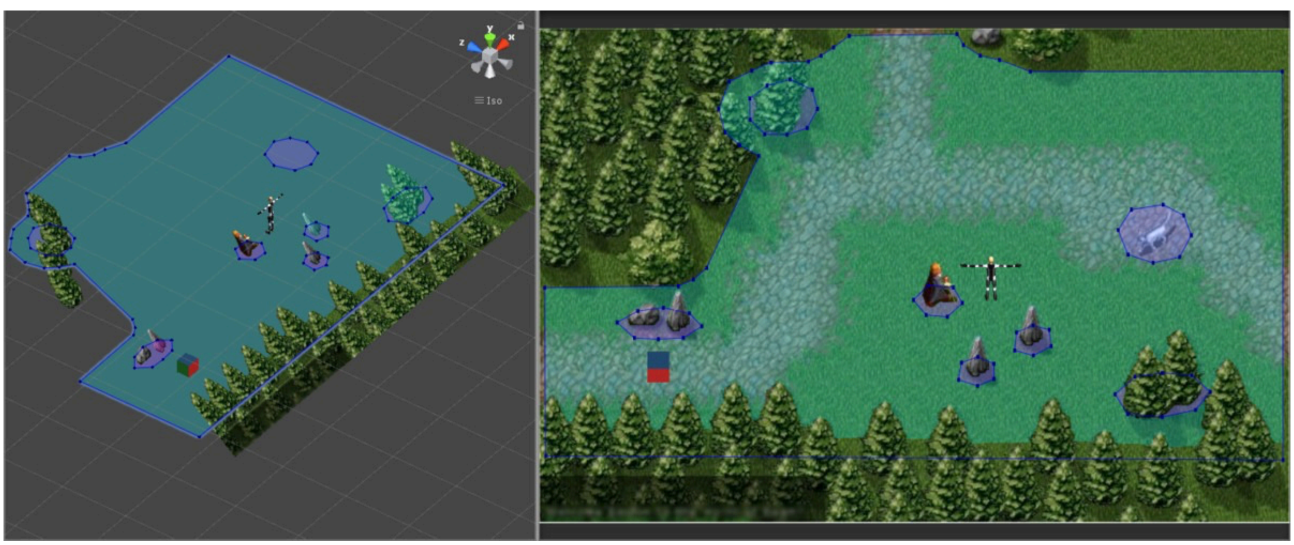
- Character models: 2 gladiator variants (distinguished by color accent, not shape), sourced from Asset Store or Blender.
- Resource nodes: Tree, Stone Block, Metal Deposit (3 simple meshes).
- Arena environment: Floor plane, colosseum walls, pillars, torches — sourced as a single low-poly arena pack if possible.
- UI elements: HP bar, resource bar, gold counter, item icons — created in Unity UI toolkit.

Crowd background characters from the original GDD are cut from the MVP. They add visual noise without contributing to gameplay and represent significant art time for zero mechanical benefit.

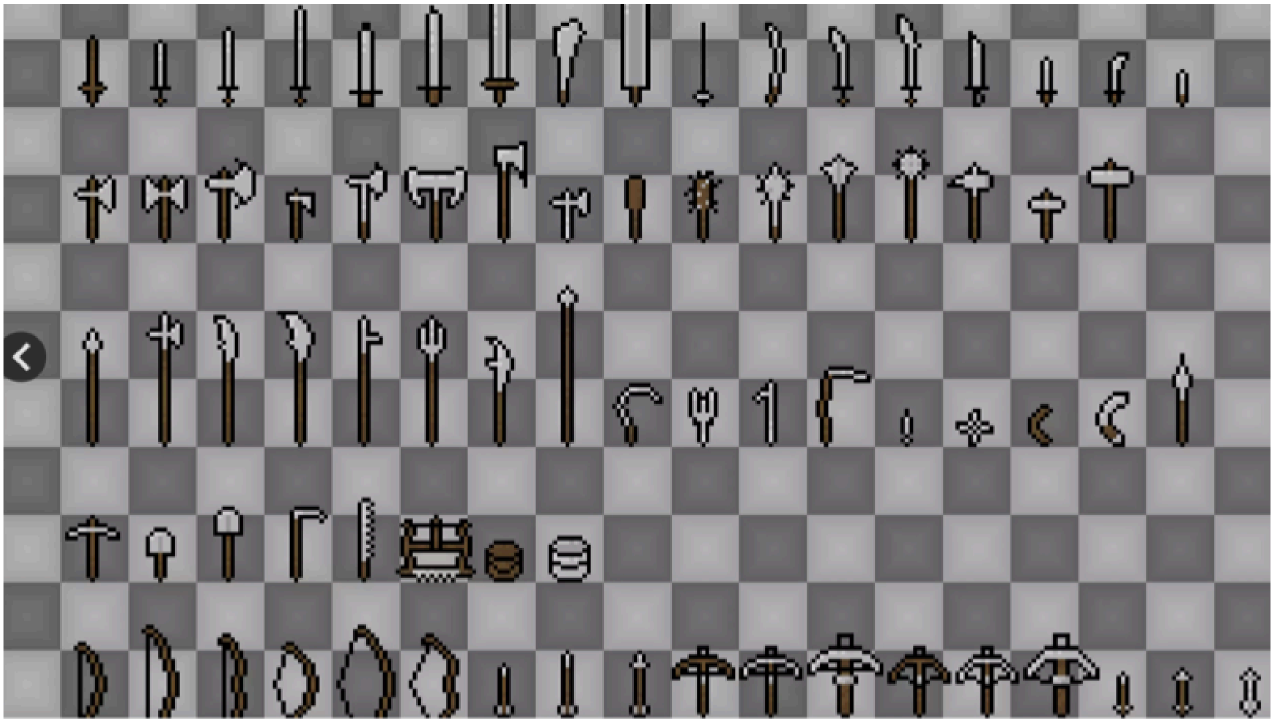
7.4 Creation Workflow

Model in Blender (or download from Asset Store) → UV unwrap → apply flat-color materials → export as FBX → import to Unity → configure prefabs and materials. Character animations (idle, walk, attack) can use Unity's built-in Animator with free animation packs (e.g., Mixamo) to further reduce custom art work.

Map (example):



Weapons (example):



8. Requirements

8.1 Must

FMR1: The system shall support 2-player local multiplayer on one keyboard using WASD for Player 1 and Arrow Keys for Player 2.

FMR2: The system shall provide destructible resource nodes with HP that players can destroy to collect resources.

FMR3: The system shall automatically convert collected resources into gold.

FMR4: The system shall display a vertical resource bar UI for each player.

FMR5: The system shall provide an item shop with at least 2 weapon options and 2 armor options.

FMR6: The system shall provide a 1-minute shopping phase.

FMR7: The system shall display a warning if a player purchases nothing during the shopping phase.

FMR8: The system shall provide a Battle Royale phase with melee combat.

FMR9: The system shall display a victory screen and provide a rematch option.

FMR10: The system shall run on a shared screen by default, with both players visible simultaneously. Split-screen is not required for the MVP (see FCR3).

NFR1: The system shall be implemented in Unity.

8.2 Should

FSR1: The system should provide at least 2–3 different resource types with different gold values.

FSR2: The system should display a phase timer visible to all players.

FSR3: The system should provide sound effects for attacks, resource collection, and phase transitions.

FSR4: The system should change the HP bar color when health is low.

8.3 Could

FCR1: The system could support up to 4 players via gamepad/controller.

FCR2: The system could provide different weapons such as spears, longswords, daggers, throwables.

FCR3: The system could support an optional split-screen mode, dividing the display into two dedicated viewports per player. This is a post-MVP enhancement; shared-screen is the default (see FMR10).

FCR4: The system could provide visual cosmetics (Hats, Glasses, ...) .

FCR5: The system could provide a session win-count tracker.

FCR6: The system could include a hidden rare node with a special item drop.

Appendix A: Glossary

MVP (Minimum Viable Product): *The initial release version of ArenaCraft containing only the core Must requirements.*

Resource Phase: *The first phase of each match (default 3 minutes) in which players collect resources from nodes and convert them to gold.*
Shopping Phase: *The second phase (1 minute) in which players spend accumulated gold on weapon and armor upgrades from the item shop.*
Battle Royale Phase: *The third and final phase in which players fight with their purchased gear until only one survivor remains.*
Resource Node: *A destructible object in the arena (Tree, Stone Block, or Metal Deposit) that players attack to harvest resources.*
Trigger Collider: *A Unity physics component used to detect overlapping objects without physical collision. Used for attack hitboxes in ArenaCraft.*
Gold: *The in-game currency automatically converted from harvested resources. Spent exclusively during the Shopping Phase.*
Low-Poly 3D: *A 3D art style using a minimal number of polygons, resulting in a faceted, stylized aesthetic. Used for all characters and environments in ArenaCraft.*
2.5D: *A visual style using 3D assets and an isometric/orthographic camera, while restricting gameplay movement to a 2D plane (X/Z axes only).*

Appendix B: Analysis Models

The following diagrams are planned to support development. Add completed diagrams here as they are produced.
B.1 Game State Machine: State diagram showing transitions between Main Menu, Resource Phase, Shopping Phase, Battle Royale Phase, and Victory Screen. [To be added]
B.2 Economy Flow Diagram: Flowchart showing how resources are harvested, converted to gold, and spent in the shop on weapons and armor. [To be added]
B.3 Component Overview: High-level diagram of the main Unity scripts and their relationships (e.g., GameManager, PlayerController, ResourceNode, ShopUI, CombatSystem). [To be added]

Appendix C: Issues List

Known open issues and design decisions that require resolution. Update this list throughout development.
C.1 Exact gold conversion values for Wood, Stone, and Metal are marked TBD in section 2.2.6. Final values must be playtested and confirmed before release.
C.2 Resource node respawn behaviour is unresolved: nodes either respawn on a timer or do not respawn at all. Decision required before Phase 1 implementation.

C.3 Attack cooldown / swing speed values for base fists and each weapon tier are not yet defined. Required for combat balancing.
C.4 Character animation source not confirmed. Mixamo is the recommended solution (see section 7.4) but licensing and import pipeline must be verified.